
Nutrition Interventions in LMICs

Automated Evidence Synthesis & Prioritization

100

Papers Reviewed

57

Full-Text (PMC)

24

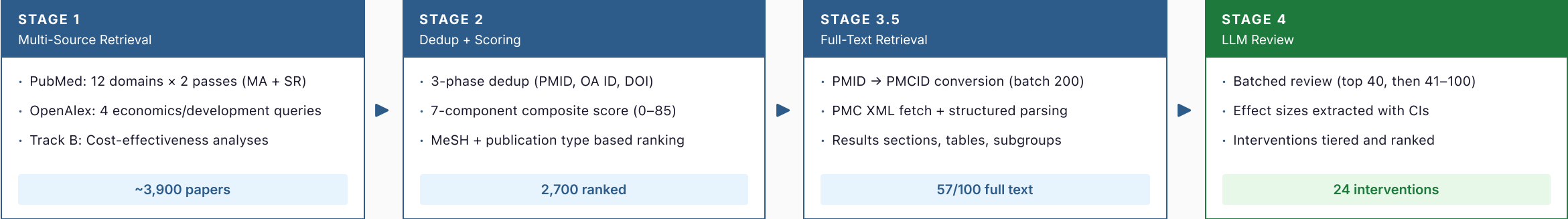
Interventions Ranked

3

Evidence Tiers

PubMed + OpenAlex systematic search · Pipeline v2.0 · May 2026

Automated Evidence Synthesis Pipeline



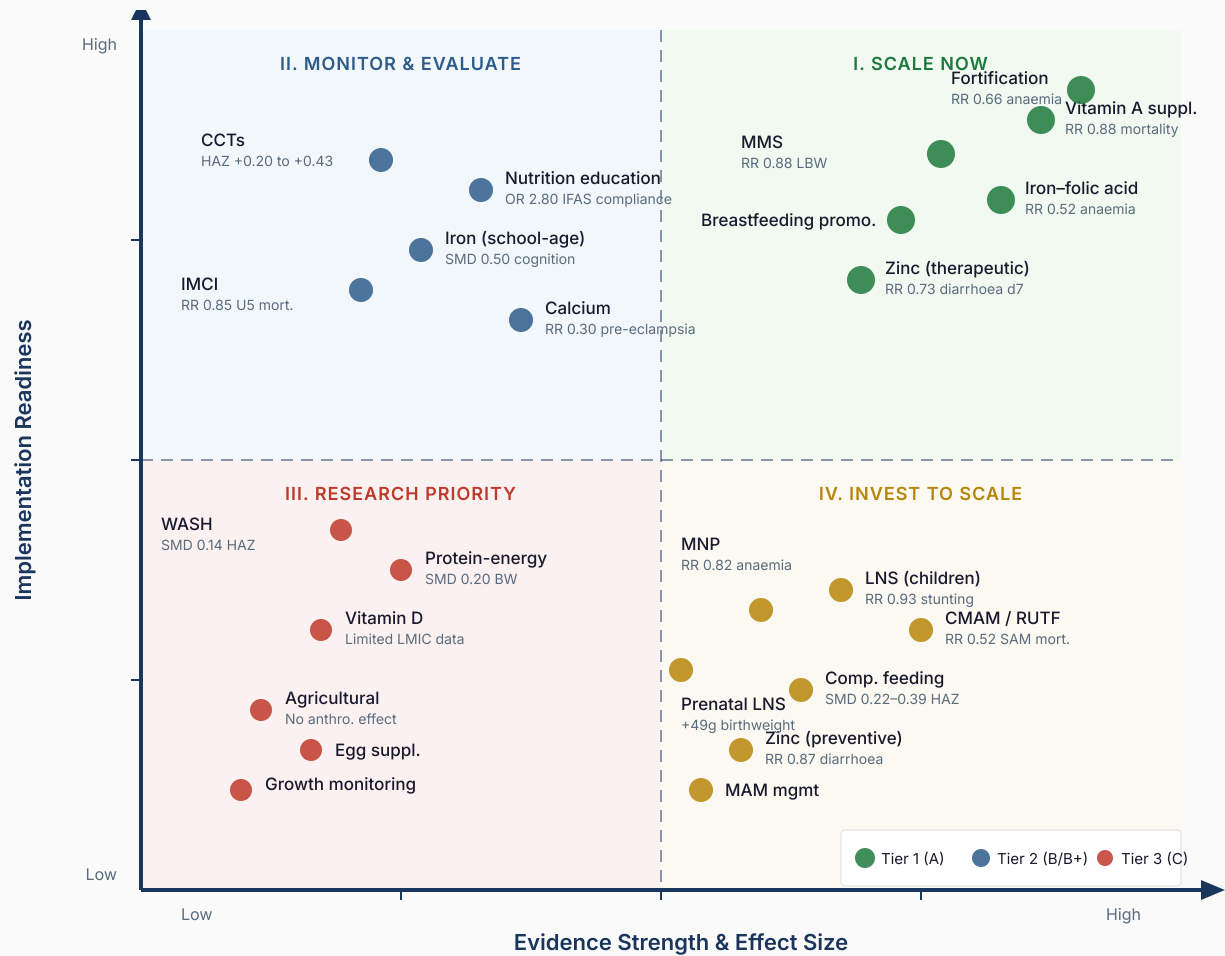
Scoring Components (max 85 points)

Study Design 0–20	Topic Relevance 0–25	Setting 0–10	Recency 0–10	Citation Impact 0–12	Open Access 0–3	Tier Bonus 0–5
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Sources: PubMed E-Utilities (MeSH indexing) · OpenAlex (economics/development) · PMC Open Access (full text) · NCBI ID Converter API

Evidence Prioritization Framework

Figure 1. Twenty-four nutrition interventions positioned by evidence strength and effect size (x-axis) against implementation readiness, a composite of cost-effectiveness and proven scalability (y-axis). Dashed lines delineate four action quadrants. Dot colour indicates evidence tier.



Quadrant Decision Rules

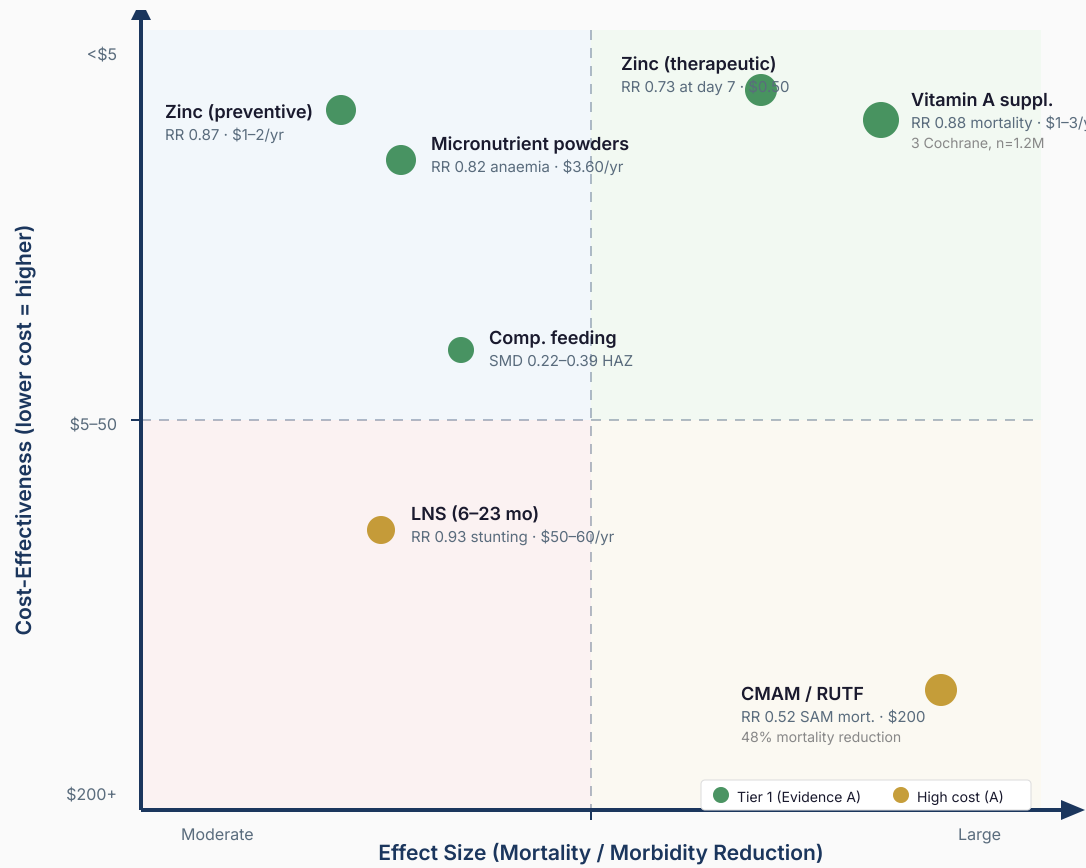
Q	Action	Criteria
I	Scale now	Evidence A + proven national + high cost-effectiveness
II	Monitor & evaluate	Good platforms, moderate evidence (B/B+)
III	Research priority	Emerging evidence (C/C+), insufficient for policy
IV	Invest to scale	Evidence A but higher cost or subnational only

Axis Composition

Axis	Composed of
Evidence (x)	GRADE certainty · # Cochrane reviews · pooled effect magnitude · consistency
Readiness (y)	Cost-effectiveness · delivery platform maturity · # countries with programmes

Children Under 5 — Intervention Prioritization

Figure 2. Child nutrition interventions (6–59 months) positioned by effect size on primary outcome (x-axis) against annual cost per child (y-axis, inverted — lower cost is higher). All interventions shown carry Evidence Rating A. Dot size proportional to evidence base breadth.



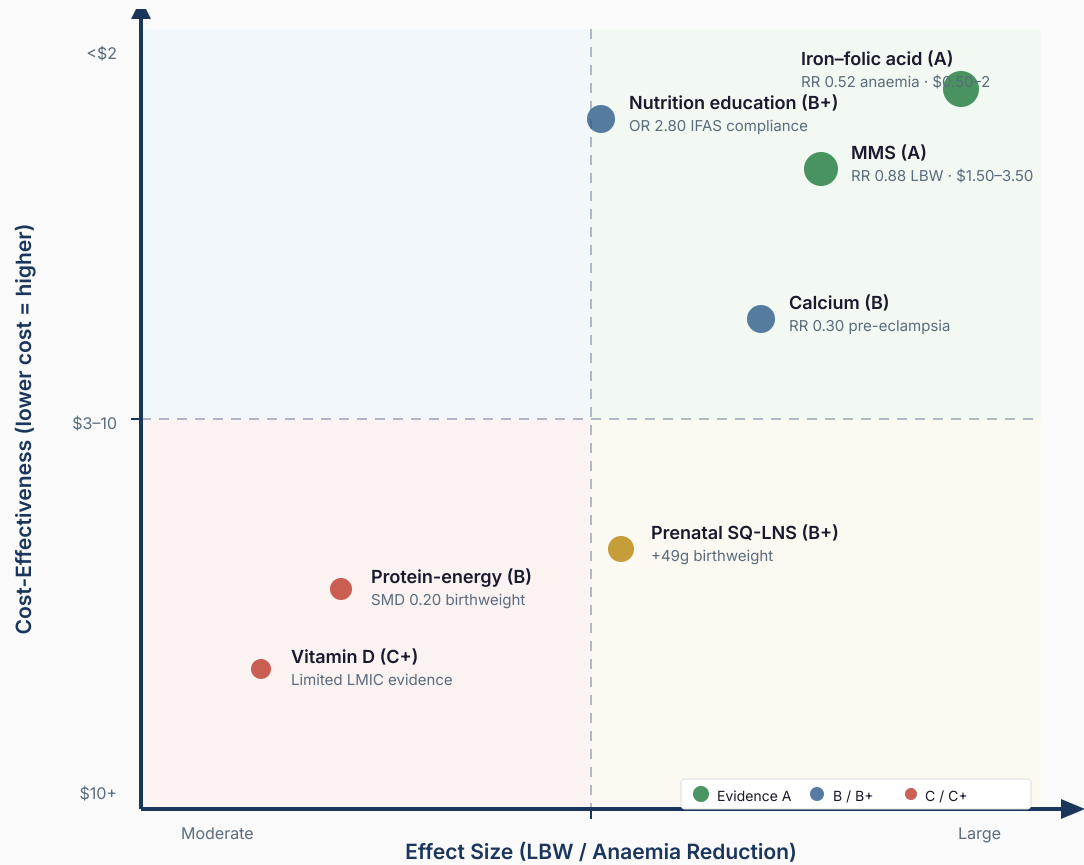
Supporting Data

Intervention	Key Effect	95% CI	Cost
VAS (6–59 mo)	RR 0.88 mortality	0.83–0.93	\$1–3/yr
Zinc (therapeutic)	RR 0.73 at day 7	0.61–0.88	\$0.50
MNP	RR 0.82 anaemia	0.76–0.90	\$3.60/yr
Zinc (preventive)	RR 0.87 diarrhoea	0.85–0.89	\$1–2/yr
Comp. feeding	SMD 0.22 HAZ	0.01–0.43	Variable
LNS (6–23 mo)	RR 0.93 stunting	0.88–0.98	\$50–60/yr
CMAM / RUTF	RR 0.52 mortality	0.43–0.64	\$200

All interventions shown carry Evidence A (multiple Cochrane reviews/meta-analyses). Higher-cost interventions (LNS, CMAM) are justified in high-burden settings where the absolute number of cases averted offsets unit cost.

Pregnant Women — Intervention Prioritization

Figure 3. Maternal nutrition interventions positioned by effect size on primary outcome (x-axis) against cost per pregnancy (y-axis, inverted). Dot colour indicates evidence rating. Evidence ratings shown in parentheses.



Key Policy Insight

MMS costs only **\$1–2 more** than IFA per pregnancy but prevents **12% more low-birthweight births** (RR 0.88 vs IFA alone). WHO conditionally recommended the IFA → MMS transition in 2020.

The Compliance Bottleneck

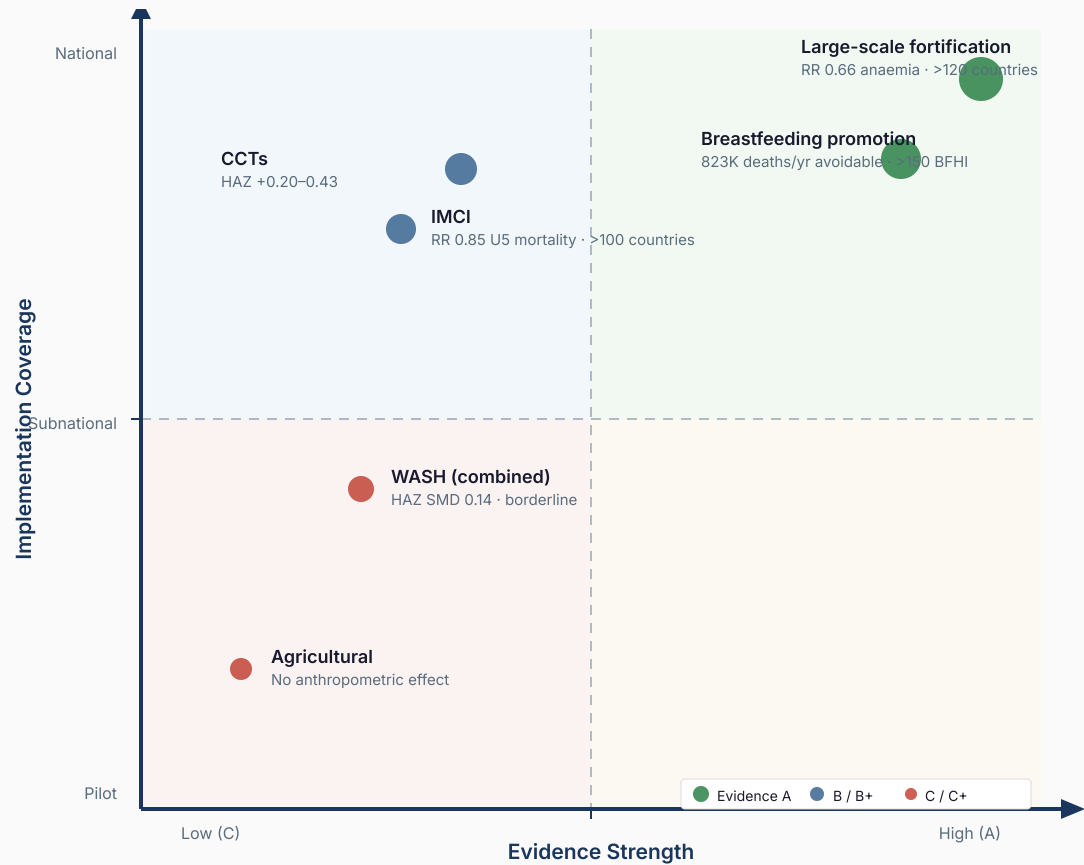
IFA supplement adherence in Sub-Saharan Africa is only **39.2%**. Nutrition education **triples adherence** (OR 2.80) — making it a force-multiplier for any supplementation programme.

Supporting Data

Intervention	Key Effect	95% CI
IFA	RR 0.52 anaemia	0.41–0.66
MMS	RR 0.88 LBW	0.85–0.91
Nutrition ed.	OR 2.80 compliance	2.04–3.83
Calcium	RR 0.30 pre-eclampsia	0.17–0.52
Prenatal LNS	+49g birthweight	IPD, 4 RCTs
Vitamin D	RR 0.40 LBW	0.23–0.71

Population-Level Interventions — Evidence vs. Reach

Figure 4. Population-level and nutrition-sensitive interventions positioned by evidence strength (x-axis) against implementation coverage across LMICs (y-axis). Dot size reflects number of countries with programmes.



Cross-Cutting Finding

76% of studies combining multisectoral policies with nutritional supplementation successfully reduced stunting, vs. variable results for single-sector approaches (PMID 31666032).

The Anaemia Paradox

Iron deficiency explains only **25% of anaemia** in preschool children — far below the commonly assumed 50%. Inflammation, malaria, helminth infections, and other micronutrient deficiencies must be addressed concurrently (PMID 27827838).

Supporting Data

Intervention	Key Effect	Countries
Fortification	RR 0.66 anaemia	>120
Breastfeeding	823K deaths/yr	>150 (BFHI)
CCTs	HAZ +0.20–0.43	Brazil, Mexico, Peru
IMCI	RR 0.85 U5 mort.	>100
WASH	HAZ SMD 0.14	Several (CLTS)
Agricultural	Diet diversity ↑	Growing

Interventions by Delivery Platform & Evidence Tier

Table 5. Classification of all 24 interventions by delivery mechanism and evidence strength. Community-based platforms concentrate the largest number of Evidence A interventions.

Delivery Platform	Evidence A (Strong)	Evidence B / B+ (Moderate)	Evidence C / C+ (Emerging)
Health Facility <i>ANC / Postnatal</i>	Iron–folic acid Multiple micronutrient suppl. Zinc for diarrhoea treatment	Calcium supplementation Nutrition education Iron suppl. (school-age) Prenatal SQ-LNS	Vitamin D supplementation
Community-Based <i>CHW Delivery</i>	Vitamin A supplementation Micronutrient powders SAM management (CMAM) Complementary feeding Breastfeeding promotion Preventive zinc suppl.	WASH interventions Community case mgmt MAM management	Growth monitoring Egg-based supplementation
Food System <i>Fortification / Production</i>	Large-scale fortification (iron, iodine, folic acid)	—	Agricultural interventions Biofortification
Cash / Social Protection	—	Conditional cash transfers IMCI strategy	—

6 of 10 Evidence A interventions are deliverable through community health worker networks — the primary scaling vehicle for child nutrition in LMICs. Health facility–based interventions concentrate in the maternal/pregnancy domain.

Vitamin A Supplementation — What the Pipeline Did Well

Pipeline Strengths

- ✓ Surfaced the **three landmark syntheses**: Imdad 2011 (CHERG/LiST) + Cochrane CD008524 (2017 & 2022 update)
- ✓ Extracted **detailed effect sizes with CIs** across mortality, morbidity, and deficiency outcomes
- ✓ Distinguished **neonatal VAS** (no benefit; possible harm in Africa) from **6–59 month VAS**
- ✓ Detected the **fixed vs random divergence**: RR 0.88 vs 0.76 for all-cause mortality
- ✓ Flagged **Asia vs Africa differential**: RR 0.69 vs 0.85 in mortality reduction
- ✓ Identified **adverse effects**: vomiting RR 1.97, bulging fontanelle in <6 mo
- ✓ Noted **declining absolute benefit** as vitamin A deficiency prevalence falls — important for programme reassessment

The Three Landmark Syntheses

Paper	Type	All -cause mortality	N children
Imdad 2011 BMC Public Health	CHERG meta-analysis (for LiST)	RR 0.75 (0.64–0.88) random	~250K
Cochrane 2017 CD008524.pub3	Systematic review + GRADE	RR 0.88 (0.83–0.93) fixed 0.76 (0.66–0.88) random	1,202,382
Cochrane 2022 CD008524 update	Update — no new RCTs	Identical to 2017	Identical

Two of the "three" are the **same Cochrane review** (2022 found no new trials); Imdad 2011 is a **CHERG meta-analysis, not Cochrane**. The fixed/random split is driven by one trial — see next slide.

Vitamin A Supplementation — Pipeline Limitations

Structural Gaps

- ✗ **Study-type misclassification:** Labelled Imdad 2011 (a CHERG meta-analysis in *BMC Public Health*) a "Cochrane review," and counted a no-new-evidence update as a third independent generation
- ✗ **Couldn't trace the divergence to its cause:** Flagged fixed 0.88 vs random 0.76 but never identified **DEVTA** as the driver (see callout)
- ✗ **Mechanism left unexamined:** The thin link is *cause-specific* mortality (diarrhoea/measles pathways underpowered), not the all-cause finding the deck dwelt on
- ✗ **CEA blind spot:** Rated "Very High" cost-effectiveness but zero cost-effectiveness papers were retrieved
- ✗ **External knowledge leak:** "\$1–3/child/yr" comes from LLM training data, not from any retrieved paper
- ✗ **Misattribution:** "823,000 deaths preventable" cited from PMID 26869575 — but that is the Lancet *Breastfeeding* Series, not VAS

The CEA Blind Spot

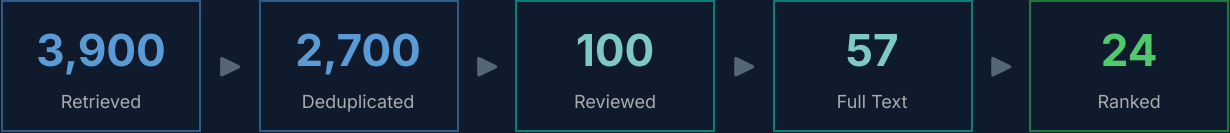
The search included a dedicated cost-effectiveness track (Track B) but retrieved **zero usable papers** for VAS. All cost claims rely on LLM external knowledge. The synthesis acknowledges this but the confidence of the "Very High" rating does not reflect the gap.

It's One Trial: DEVTA

Fixed-effect RR 0.88 vs random-effects RR 0.76 isn't an abstract "model choice" — it's **DEVTA** (India, ~1M children, RR ~0.96), which carries **61.7% of the fixed-effect weight** and drags the pooled estimate up. Down-weight it (random-effects) and you recover Imdad 2011's ~24% reduction. A domain expert names DEVTA in one sentence; the LLM produced only a vague "models differ."

A mix of LLM reasoning gaps (study-type misclassification, missing DEVTA, mechanism left unexamined) and structural pipeline gaps (no CEA source, no trial-overlap detection). Fixes need both better prompting and pipeline changes — see notes.

Summary & Key Takeaways



Top 5 "Scale Now" Interventions

#	Intervention	Key Effect (95% CI)	Cost	Evidence
1	Vitamin A suppl. (6–59 mo)	RR 0.88 all-cause mortality (0.83–0.93)	\$1–3/child/yr	A — 3 Cochrane
2	Iron–folic acid (pregnancy)	RR 0.52 maternal anaemia (0.41–0.66)	\$0.50–2/pregnancy	A
3	Multiple micronutrient suppl.	RR 0.88 low birthweight (0.85–0.91)	\$1.50–3.50/pregnancy	A — Cochrane
4	Large-scale fortification	RR 0.66 anaemia (0.59–0.74)	\$0.05–0.50/person/yr	A
5	Exclusive breastfeeding promo.	823K under-5 deaths preventable/yr	Very low per beneficiary	A

Key Limitations

- ▶ **No cost-effectiveness papers retrieved** despite dedicated search track — all cost ratings rely on LLM external knowledge
- ▶ **43/100 papers abstract-only** — effect sizes not independently verified from full text for nearly half the evidence base
- ▶ **Geographic concentration** in South Asia and East/West Africa — Latin America, Central Asia, Pacific Islands underrepresented